Step-by-Step Guide: Installing Aruba ClearPass 6.12 on Proxmox VE 8.3.4

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Introduction

Aruba ClearPass is a powerful **network access control (NAC) and policy management solution** that allows organizations to enforce security policies for devices accessing their network.

📌 ***Note: ClearPass Installation on Proxmox VE is not supported officially, so this guide is only for lab purposes.***



In this guide, we will walk through the **installation of Aruba ClearPass 6.12** on **Proxmox VE** 8.3 using a **KVM RAW image**.

This tutorial covers:

- ✅ **Downloading the image**
- ✅ **Creating a virtual machine in Proxmox**
- ✅ **Uploading and writing the ClearPass RAW image to disk**
- ✅ **Configuring the virtual machine and network settings**
- ✅ **Booting and configuring ClearPass for first-time use**

By the end of this guide, you'll have a **fully functioning ClearPass instance** running in **Proxmox VE for Lab use.**

Step 1: Download the ClearPass 6.12 KVM RAW Image

Before we begin, we need to download the **Aruba ClearPass 6.12 KVM RAW image**.

1. Go to the HPE Networking Support Portal

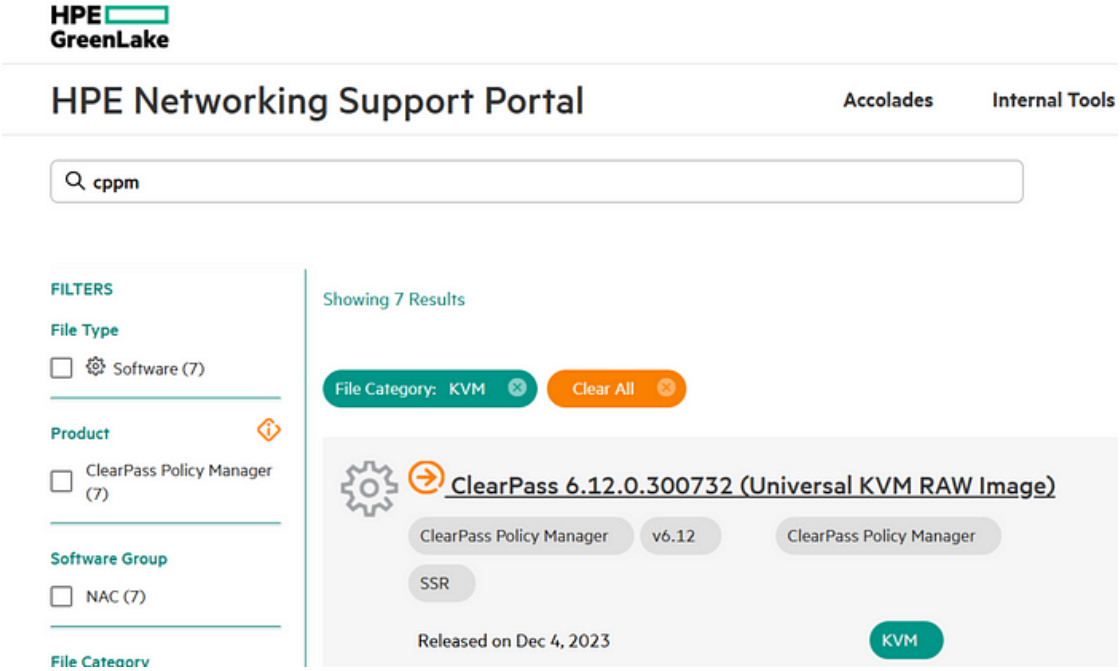
- Open a **web browser** and navigate to **HPE Networking Support Portal** (https://networkingsupport.hpe.com/home).
- **Log in with your HPE account** (required for access).

2. Search for "CPPM"

- In the **search bar**, type **CPPM** and press **Enter**.
- Apply the following **filters**:
- **Product:** ClearPass Policy Manager
- **File Category:** KVM

3. Download the RAW Image

- Locate **"ClearPass 6.12.0.300732 (Universal KVM RAW Image)" or later version**.
- Click the **download link** and save it to your **local machine**.



📌 **Tip:** This file is large, so ensure you have a stable internet connection.

✅ **Now, you have the ClearPass 6.12 image ready for deployment.**

Step 2: Create a Virtual Machine in Proxmox

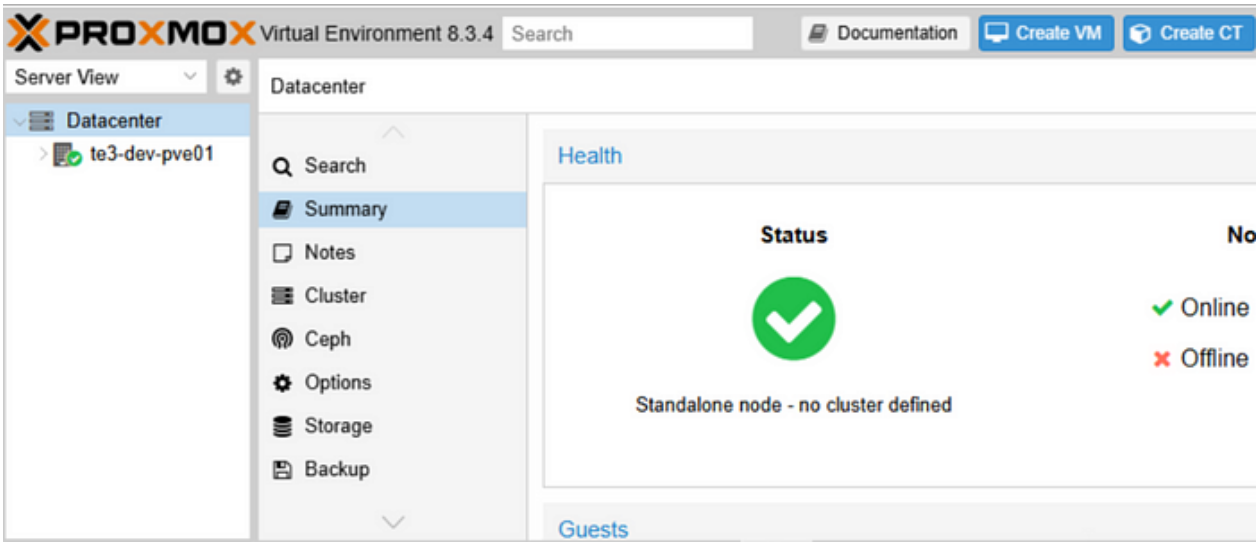
Now, let's create a new **VM for ClearPass** in Proxmox.

1. Log in to Proxmox

- Open your **Proxmox Web UI**.

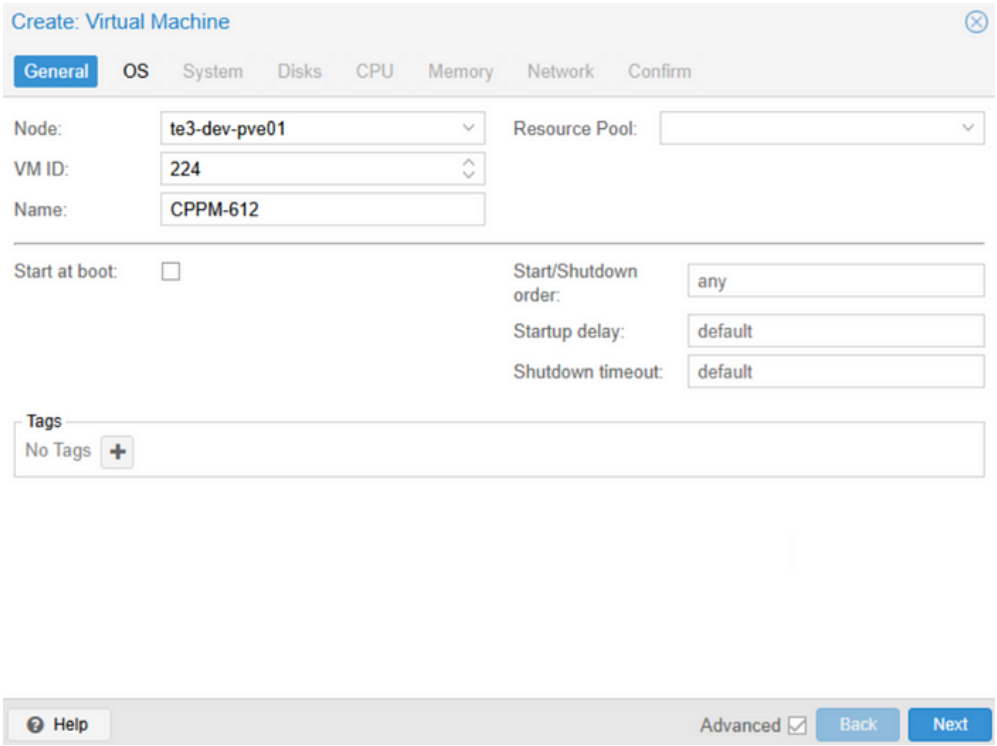
2. Creating a New Virtual Machine

- Click **“Create VM”** in the top-right corner.



3. General Settings

- **Node:** Select your Proxmox node.
- **VM ID:** Choose a unique ID.(eg.224 in this case)
- **Name:** Set the VM name (eg. **CPPM-612**.)
- **Start at Boot:** (Optional) Leave **unchecked** for now.
- Click **Next**.

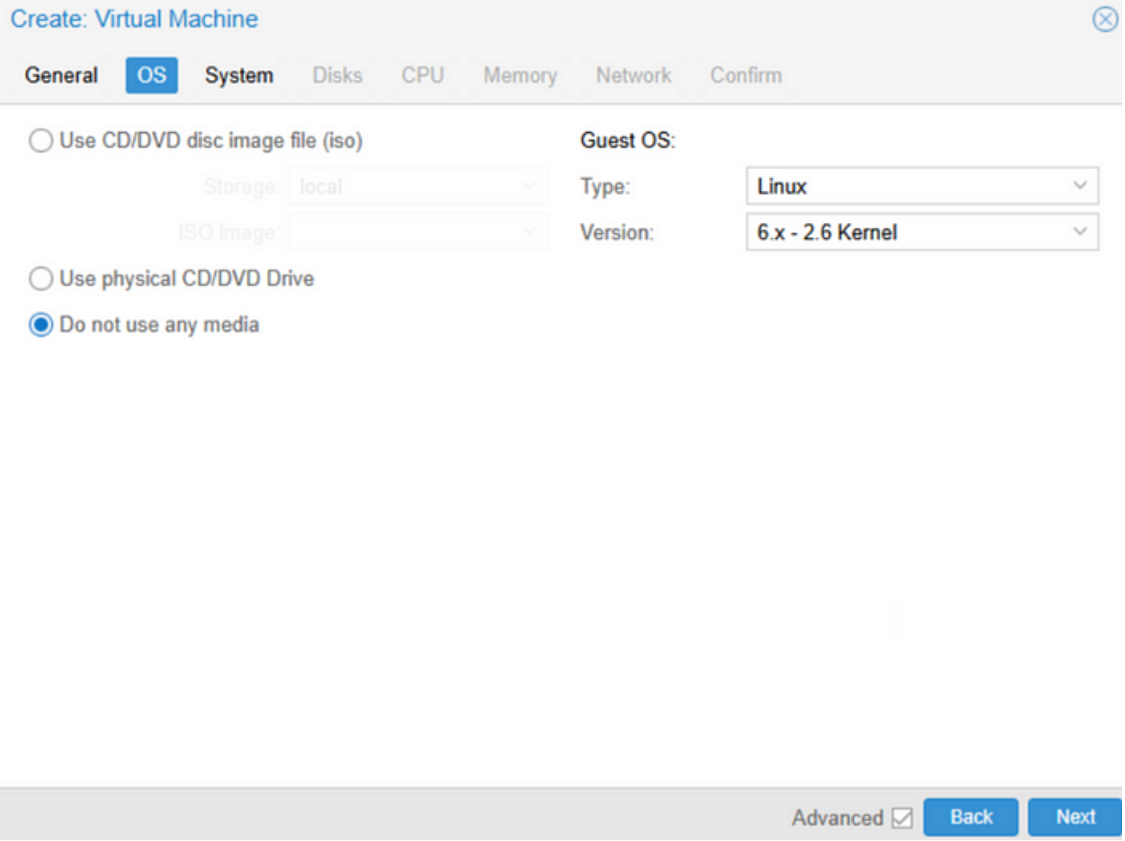


✅ **Now, the VM has been created and is ready for configuration.**

Step 3: Configure the Virtual Machine Settings

1. OS Settings

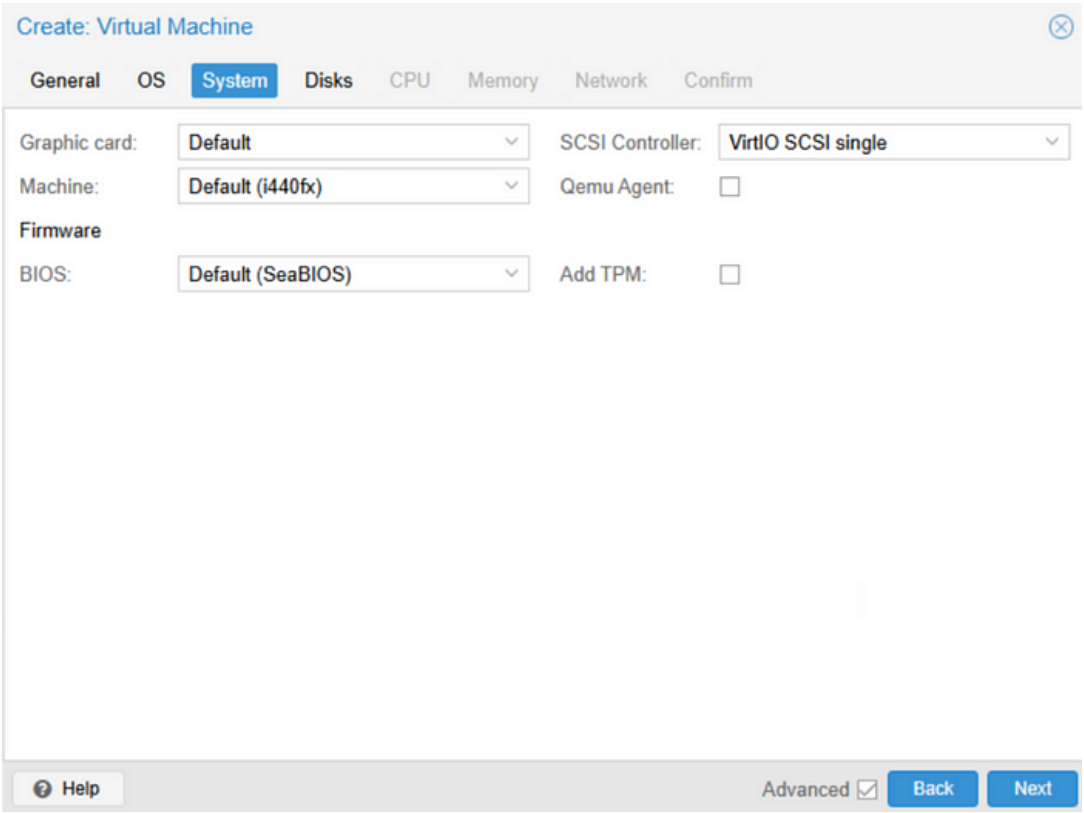
- **Guest OS Type: Linux**
- **Version: 6.x — 2.6 Kernel**
- **Installation Media: Do not use any media**
- Click **Next**.



📌 **Note:** This setting ensures we manually install the ClearPass image later.

2. System Settings

- **Graphics Card: Default**
- **Machine Type: Default (i440fx)**
- **BIOS: Default (SeaBIOS)**
- **SCSI Controller: VirtIO SCSI single** (not used but keep default)
- Click **Next**.

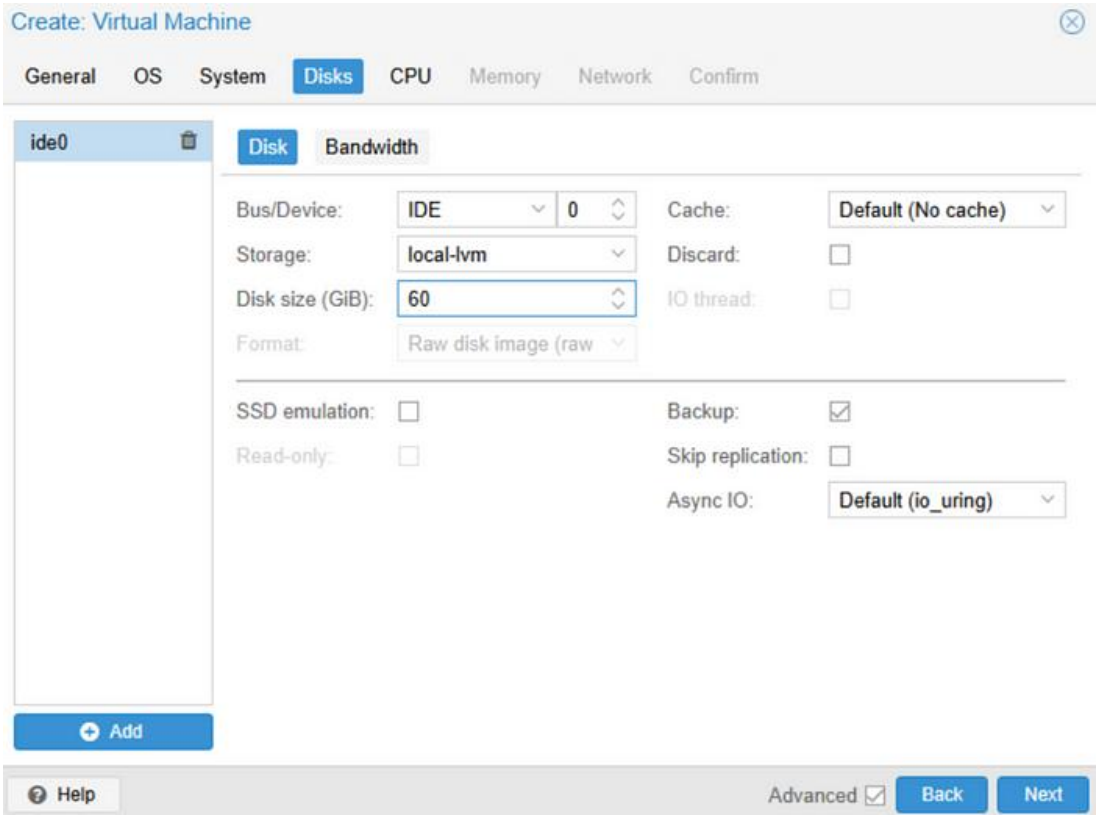


✅ **Now, the system is configured correctly.**

Step 4: Add Virtual Disks

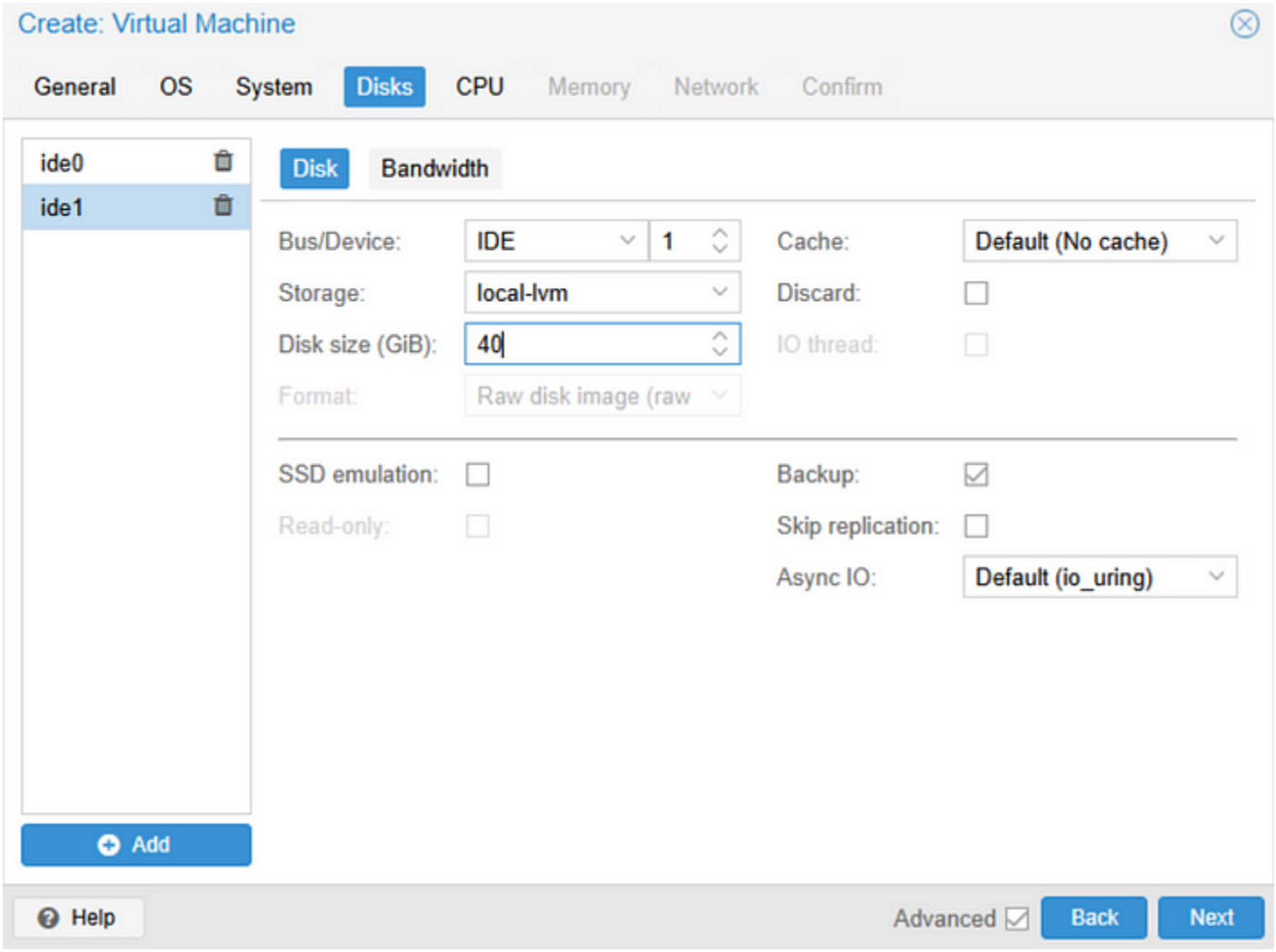
1. First Disk (System Disk)

- **Bus/Device: IDE 0**
- **Storage: local-lvm**
- **Disk Size: 60GB (Minimum required)**
- **Cache: Default (No cache)**
- Click **Next**.



2. Second Disk (Data Disk)

- Click **“Add”** to create an additional disk.
- **Bus/Device: IDE 1**
- **Storage: local-lvm**
- **Disk Size: 40GB (Minimum required)**
- **Cache: Default (No cache)**
- Click **Next**.

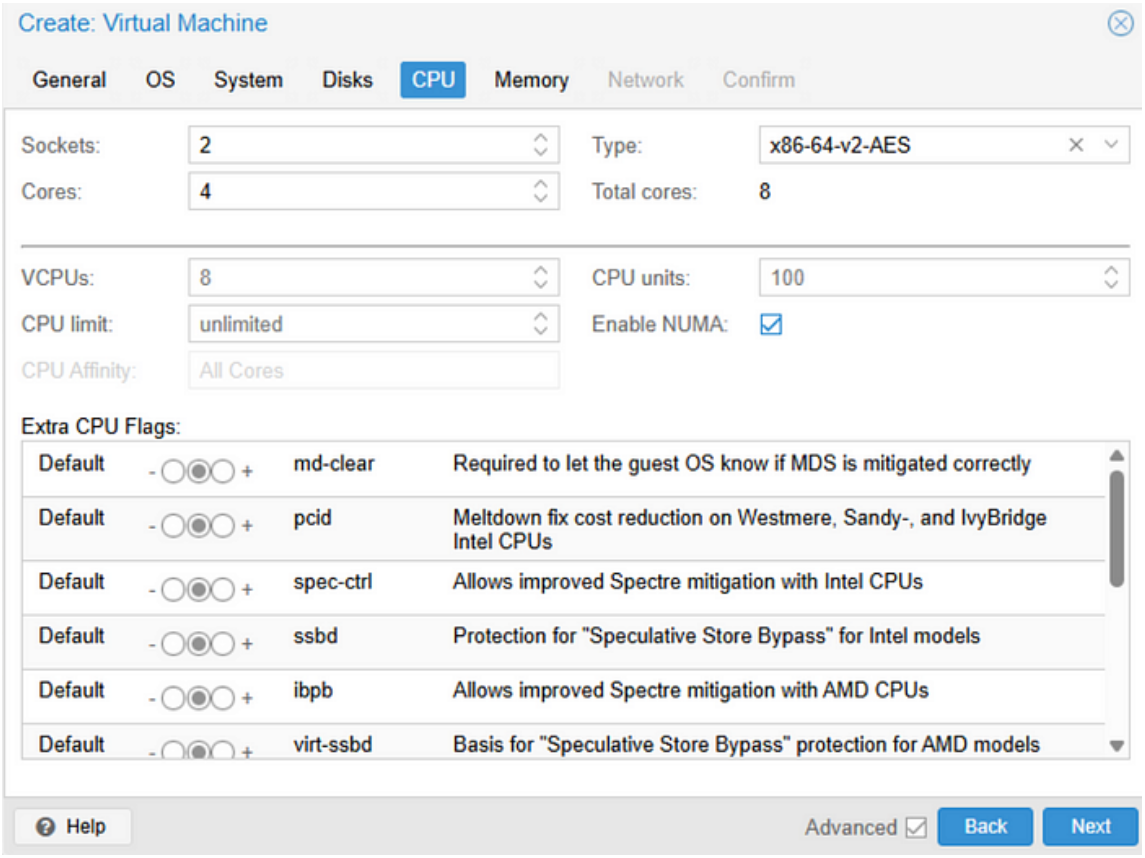


✅ **The VM now has two disks — one for the system and another for data. (Tips: Do not use SCSI or SATA disk, otherwise you will encounter UUID issues.)**

Step 5: Configure CPU & Memory

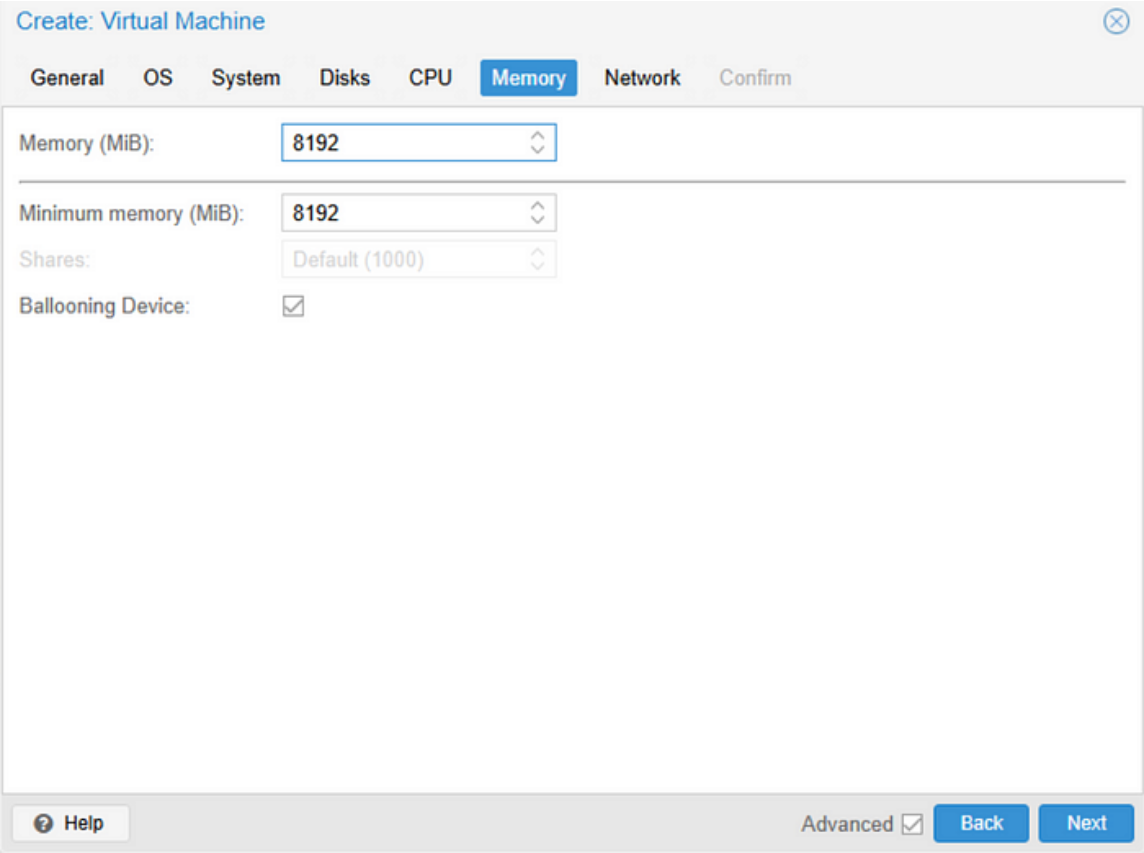
1. CPU Settings

- **Sockets: 2**
- **Cores per Socket: 4** (Total **8 cores**)
- **Type: x86–64-v2-AES**
- **VCPUs: 8**
- **Enable NUMA:** Checked
- Click **Next**.



2. Memory Settings

- **Memory (MiB): 8192 MiB (8GB minimum, 12GB recommended)**
- Click **Next**.



📌 **Why 8GB+ RAM?**

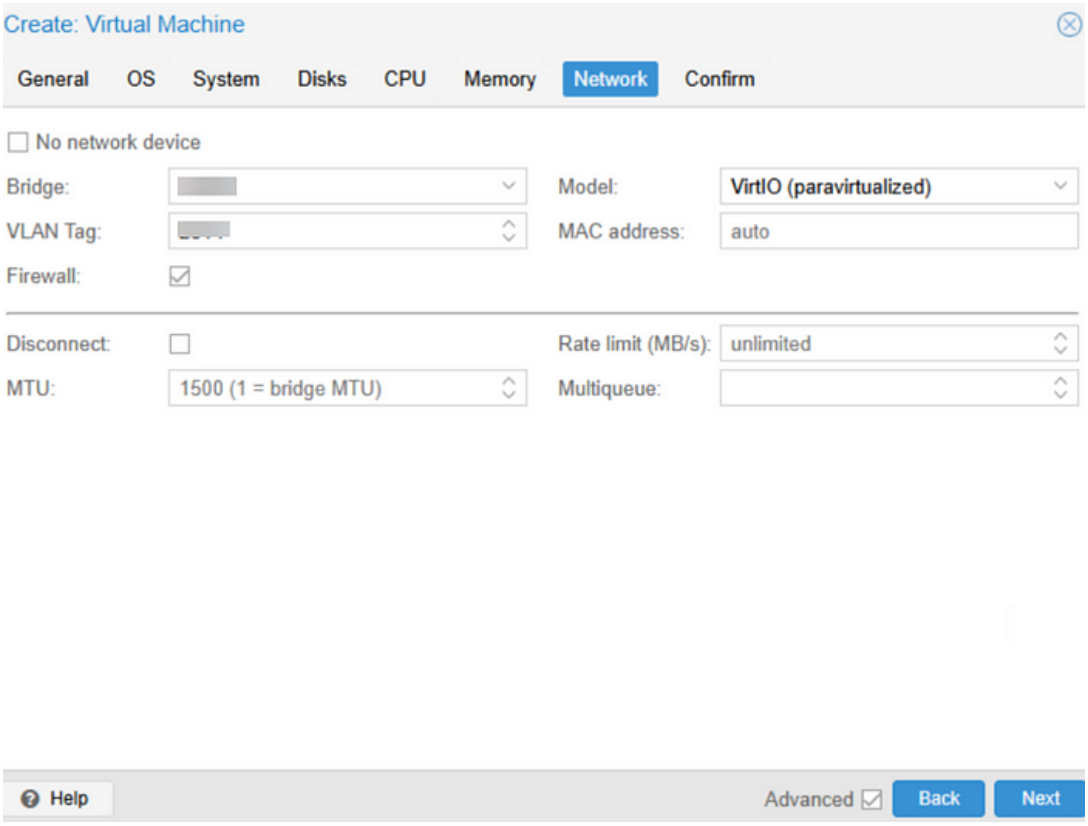
- Aruba ClearPass requires significant memory for **policy processing and logging**. 8GB is minimum for lab purposes, I prefer to add it to 12GB.

✅ **Now, the VM is configured with CPU and memory.**

Step 6: Configure Network Interfaces

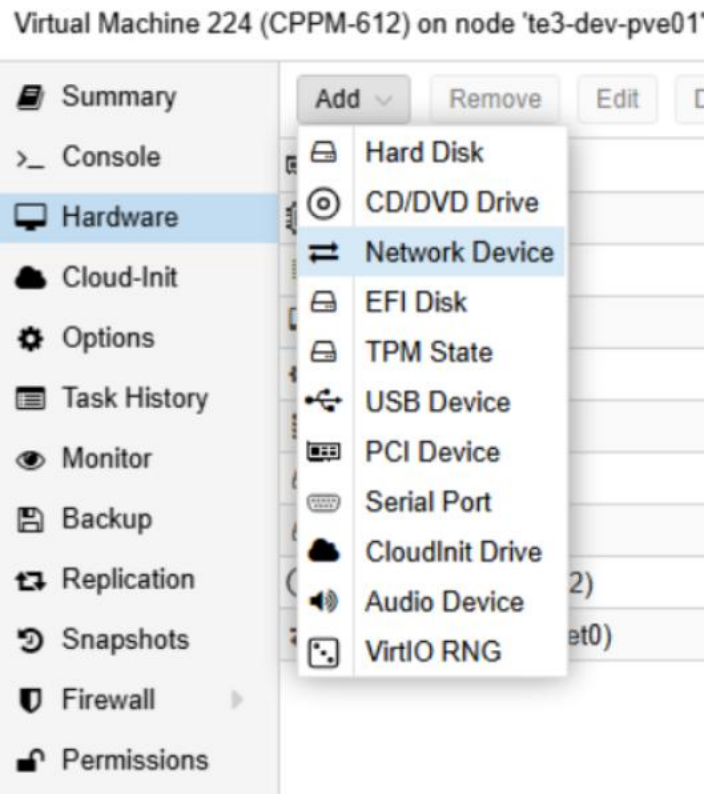
1. Management Network (First Port)

- **Bridge: <yours>**
- **Model: VirtIO (paravirtualized)**
- **VLAN Tag: <yours>** *(Varies by setup)*
- **Firewall: Checked**
- Click **Next** and **Confirm** the configuration.



2. Add extra Data Network (Second Port)

- **Go to "Hardware" > "Add" > "Network Device"**.
- **Bridge: <yours>**
- **Model: VirtIO (paravirtualized)**
- **VLAN Tag: <yours>** *(Varies by setup,* ***also you may want to check extra tips in the end about the network interface order based on mac address)***
- Click **Add**.



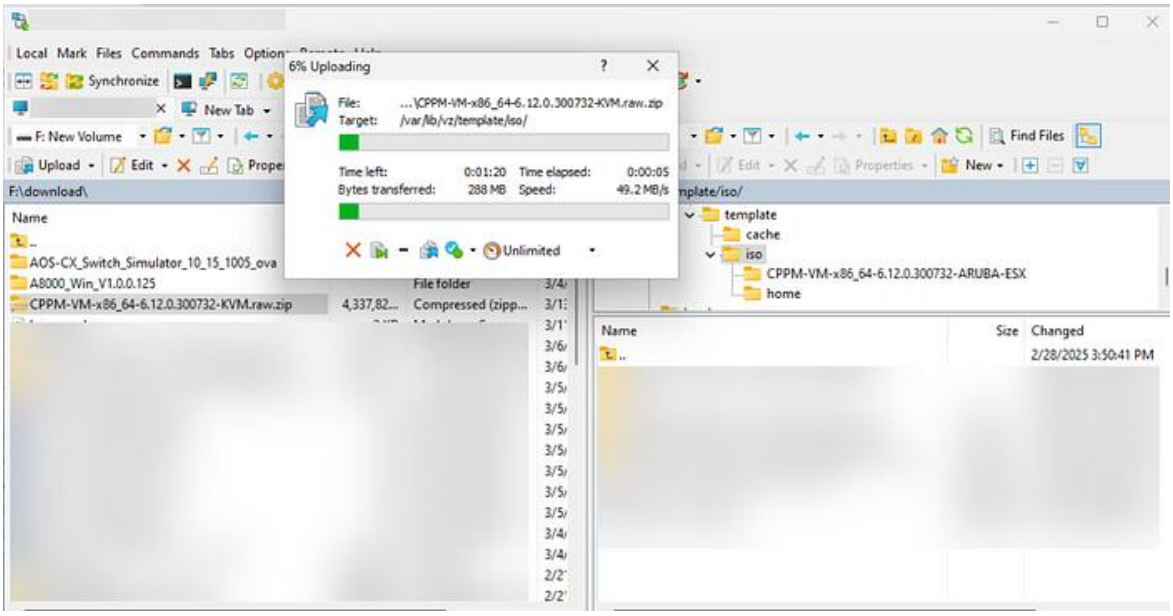
📌 **Why two network interfaces?**

- **Management Port:** Used for **admin access and policy configuration**.
- **Data Port:** Used for **client authentication and RADIUS traffic**.

Step 7: Upload & Extract the ClearPass RAW Image

1. Upload the ZIP File to Proxmox

- Open **WinSCP** and log in to your **Proxmox server**.
- **Upload the file** to: /var/lib/vz/template/iso



2. Extract the Image

- **SSH into Proxmox** and run:

cd /var/lib/vz/template/iso

unzip CPPM-VM-x86_64–6.12.0.300732-KVM.raw.zip

```
root@te3-dev-pve01:/var/lib/vz/template/iso# unzip CPPM-VM-x86_64-6.12.0.300732-KVM.raw.zip
Archive:  CPPM-VM-x86_64-6.12.0.300732-KVM.raw.zip
  inflating: home/jenkins/KVMImage/CPPM-VM-x86_64-6.12.0.300732-KVM.raw
root@te3-dev-pve01:/var/lib/vz/template/iso#
```

- The RAW image will be saved at: ***home/jenkins/KVMImage/CPPM-VM-x86_64–6.12.0.300732-KVM.raw***

📌 **Ensure the RAW image is extracted correctly before proceeding.**

✅ **Now, the ClearPass image is extracted and ready to be written.**

Step 8: Write the RAW Image to the VM Disk

Now that we've uploaded and extracted the **ClearPass 6.12 RAW image**, we need to **write it to the system disk** using dd.

1. Identify the Logical Volume (LV) for the VM

We need to **find the correct disk** to avoid overwriting the wrong VM.

- SSH into **Proxmox** and run: ***lvdisplay***
- Look for the **logical volume (LV) path** associated with your ClearPass VM (e.g., vm-224-disk-0 for VM ID 224).
- Identify the **block device number (dm-ID)** from the **Block device** field.
- Example output:

```
  --- Logical volume ---
  LV Path                /dev/pve/vm-224-disk-0
  LV Name                vm-224-disk-0
  VG Name                pve
  LV UUID                ZyifXD-IOb6-BI2f-jsf0-WyqN-rglo-CSDuK0
  LV Write Access        read/write
  LV Creation host, time te3-dev-pve01, 2025-03-13 15:37:52 -0400
  LV Pool name           data
  LV Status              available
  # open                 0
  LV Size                60.00 GiB
  Mapped size            0.00%
  Current LE             15360
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     768
  Block device           252:75

  --- Logical volume ---
  LV Path                /dev/pve/vm-224-disk-1
  LV Name                vm-224-disk-1
  VG Name                pve
  LV UUID                ygXNxM-ew1j-BJEK-Ofzy-SZq9-Zdy0-s3JUAq
  LV Write Access        read/write
  LV Creation host, time te3-dev-pve01, 2025-03-13 15:37:52 -0400
  LV Pool name           data
  LV Status              available
  # open                 0
  LV Size                30.00 GiB
  Mapped size            0.00%
  Current LE             7680
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     768
  Block device           252:76

root@te3-dev-pve01:/var/lib/vz/template/iso# lvdisplay
```

In this case, the block device is 252:75 for the disk 0, meaning the device path is /dev/dm-75.

📌 **Be extra careful — writing to the wrong device will destroy another VM!**

2. Write the RAW Image to Disk

- Run the dd command to copy the image:

dd if=/var/lib/vz/template/iso/home/jenkins/KVMImage/CPPM-VM-x86_64–6.12.0.300732-KVM.raw of=/dev/dm-<Your dm id>

- This process will take some time depending on **disk speed**.
- Example output:

```
root@te3-dev-pve01:/var/lib/vz/template/iso# dd if=/var/lib/vz/template/iso/home/jenkins/KVMImage/CPPM-VM
-x86_64-6.12.0.300732-KVM.raw of=/dev/dm-75

94371840+0 records in
94371840+0 records out
48318382080 bytes (48 GB, 45 GiB) copied, 329.859 s, 146 MB/s
```

✅ **Now, the ClearPass image has been successfully written to the VM disk!**

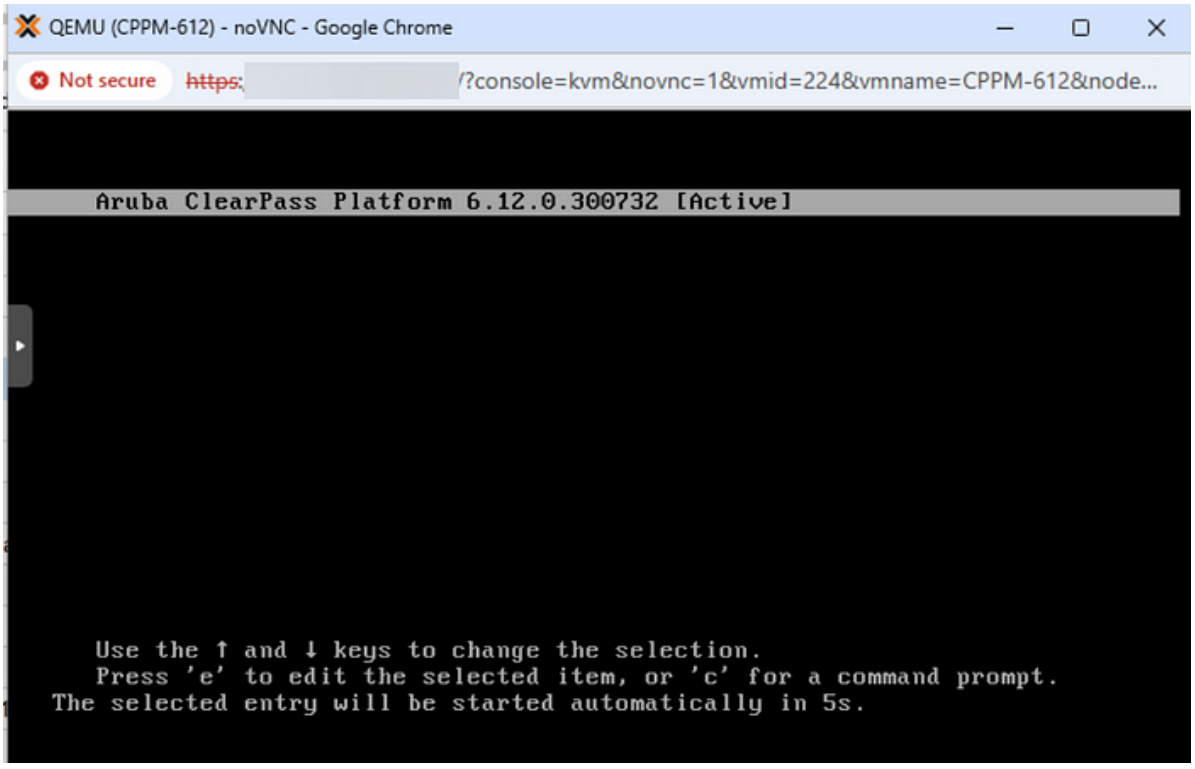
Step 9: Start the VM and Configure ClearPass

1. Start the VM

- In **Proxmox**, select the **ClearPass VM**.
- Click **"Start"**.
- Open the **Console** to monitor the boot process.

2. ClearPass Boot Menu

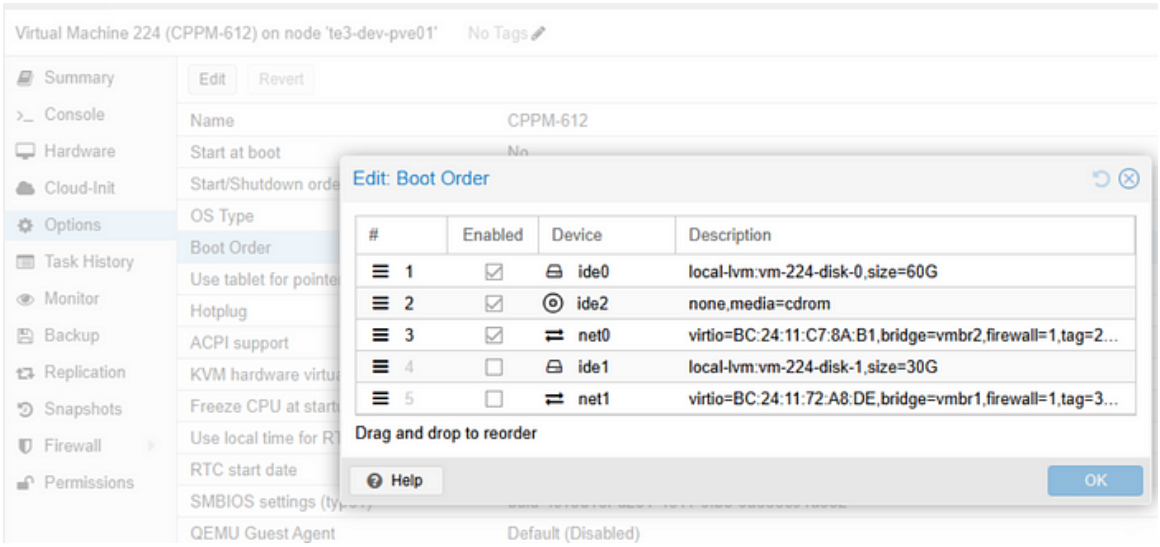
- You will see the **Aruba ClearPass 6.12 boot menu**:



- Let the system boot automatically.

If the system doesn't boot, ensure that IDE 0 is set as the first boot device.

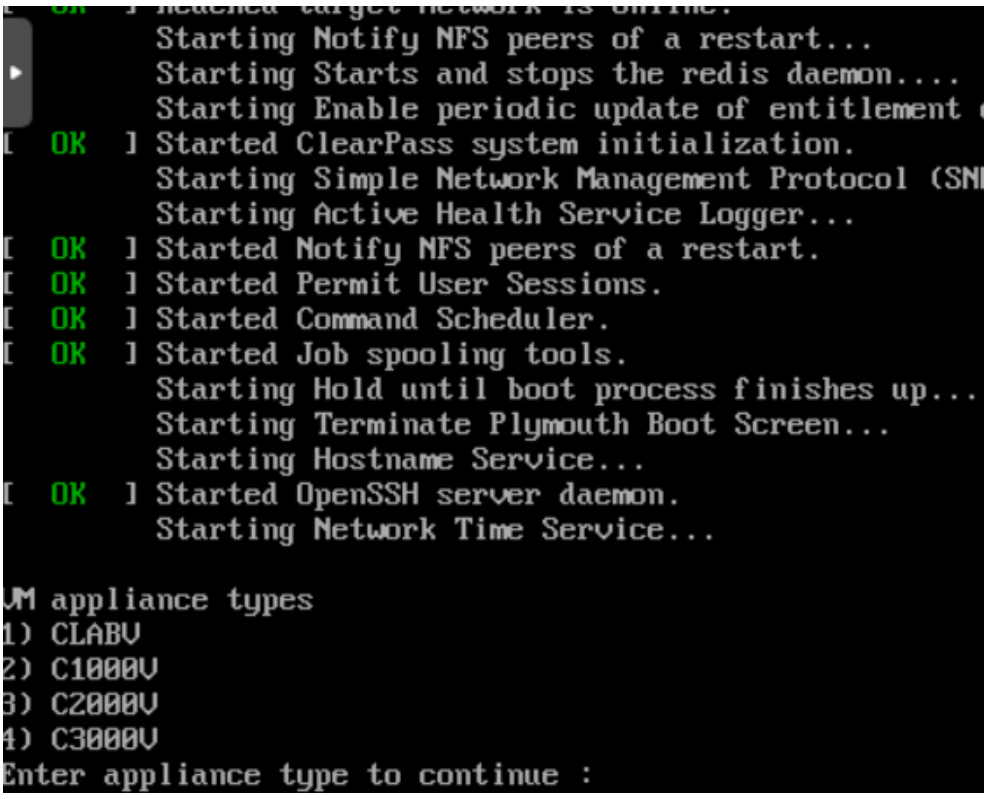
- **Go to:** Options → Boot Order
- Ensure **IDE0 (60GB system disk) is first**.



Step 10: Initial ClearPass Setup

1. Select Appliance Type

- The system will prompt:

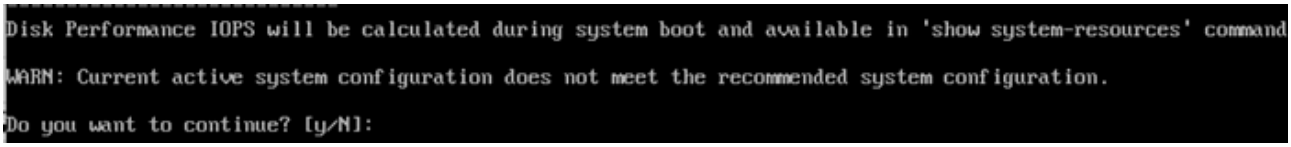


Enter appliance type to continue:

- Type **1** (**CLABV**) for **lab use**, then press **Enter**.

2. Ignore System Configuration Warnings

- The system may warn about memory and disk size: **WARN: Current active system configuration does not meet the recommended system configuration.** Since it is only for lab use, ignore this warning and continue.

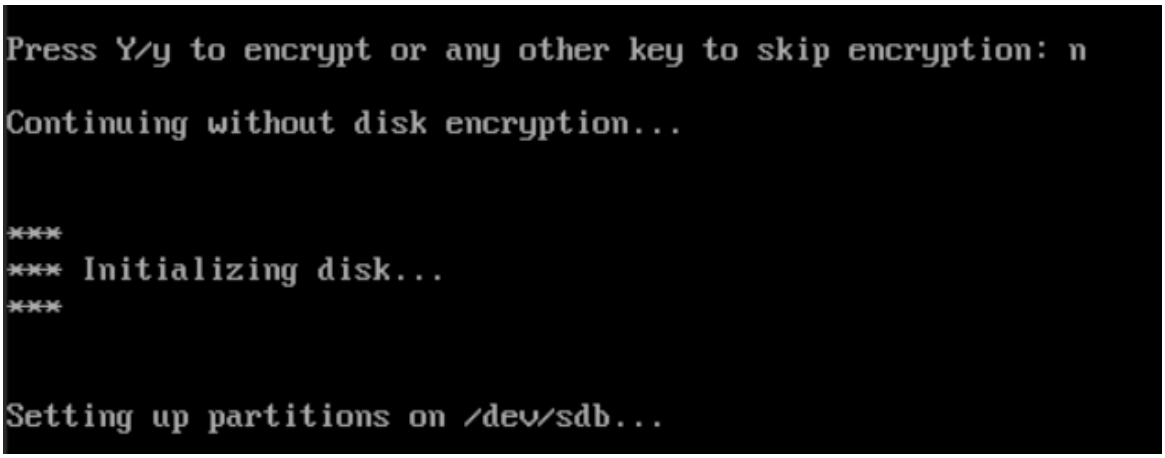


Do you want to continue? [y/N]:

- Type **y** and press **Enter**.

3. Disk Initialization & Encryption

- The system will ask if you want to encrypt the disk:



- **Type n** to skip encryption and press **Enter**.

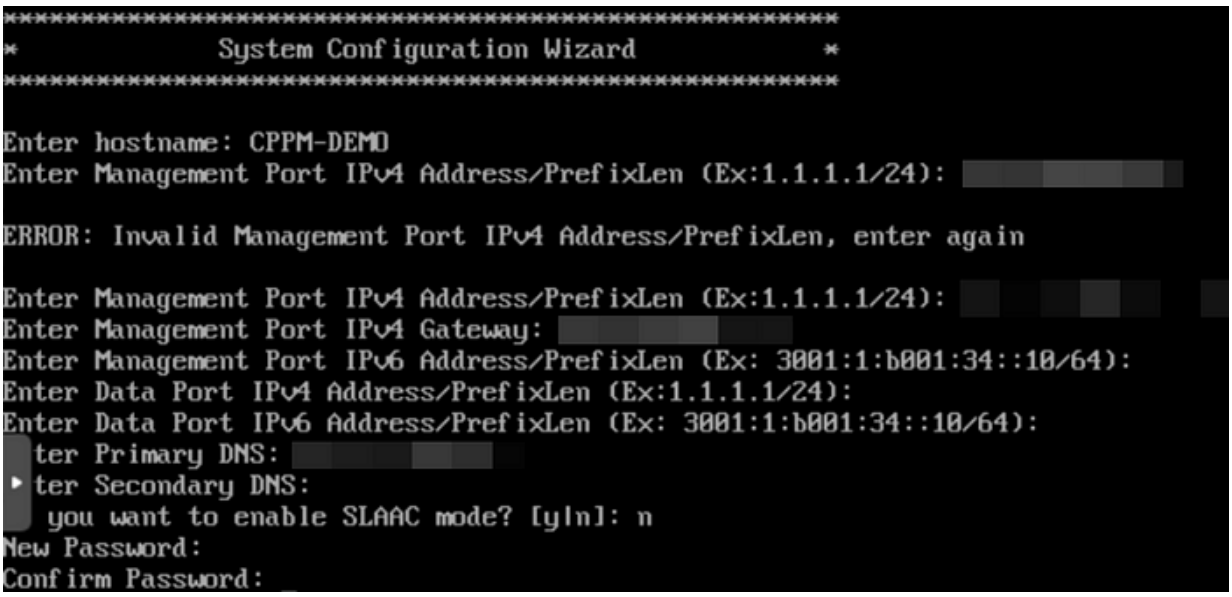
📌 **Encryption is not required for lab use** and may increase installation time.

Step 11: Configure Networking

After initialization, the **System Configuration Wizard** starts.

1. Set Hostname & Management Network

- Enter **Hostname** (e.g., CPPM-DEMO).
- Enter **Management Port IPv4 Address**
- Enter **Management Port Gateway**



📌 **If you see an "Invalid Management Port IPv4" error, ensure you include /24 in the subnet mask.**

2. Set Primary & Secondary DNS

- Primary DNS: **<Your DNS server>**
- Secondary DNS: *(Optional, leave blank and press Enter)*

3. Enable/Disable SLAAC for IPv6

- When prompted: Do you want to enable SLAAC mode? [y/n]: n
- **Type n** and press **Enter**.

4. Set the Admin Password

- Enter a **new password** for ***appadmin*** user.
- **Confirm the password** and press **Enter**.

Step 12: Final Configuration

1. System Date & Time Settings (Optional)

- The system will ask: Do you want to configure system date time information? [y/n]: n
- **Type n** and press **Enter**. (Choose Y you need to configure NTP server etc)

2. Enable/Disable FIPS Mode

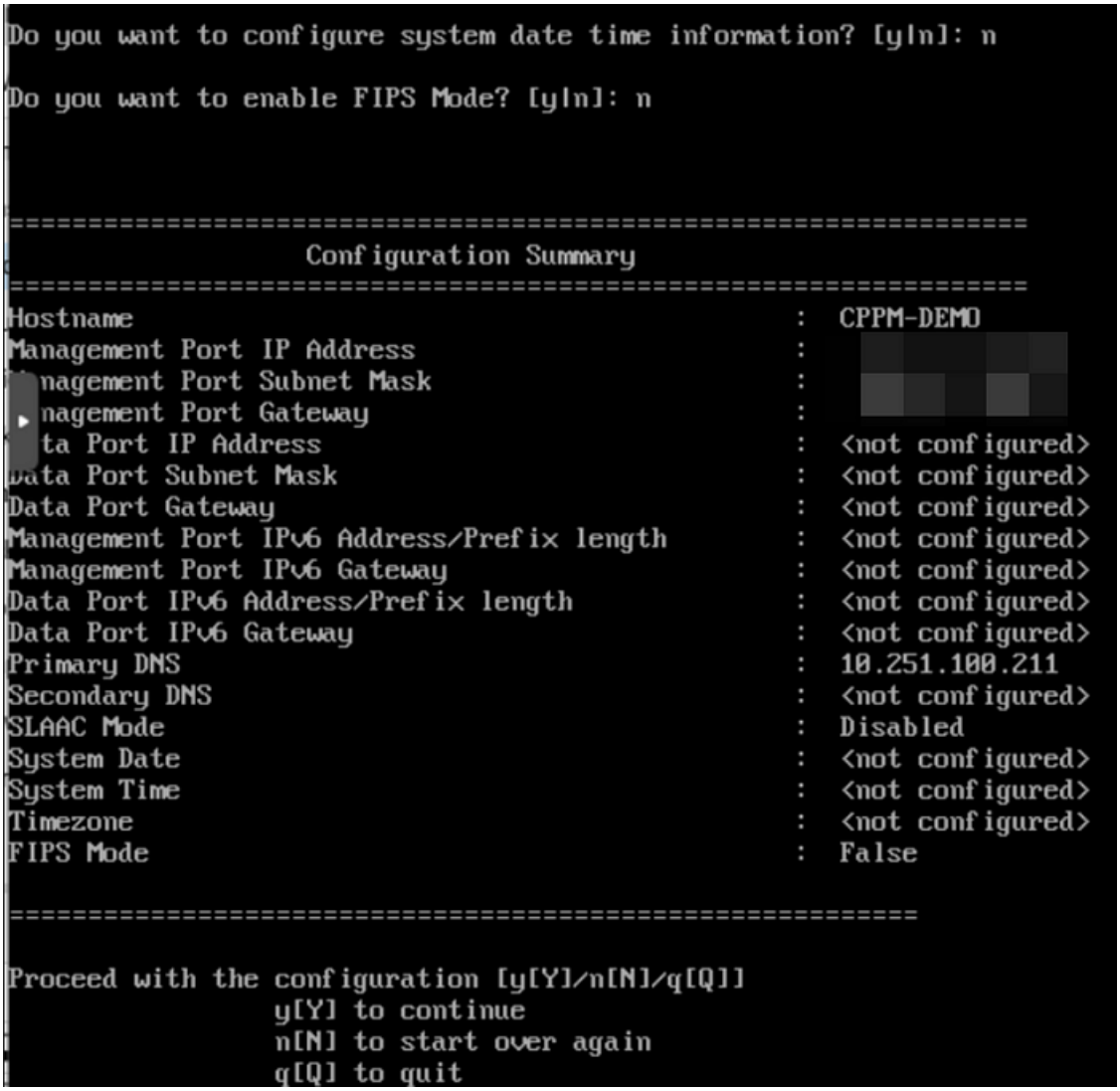
- The system will ask: Do you want to enable FIPS Mode? [y/n]: n
- **Type n** and press **Enter**.

📌 **FIPS (Federal Information Processing Standards) Mode is not required for most lab environments.**

3. Review Configuration

- The system will display the **final configuration summary**.

- If everything looks correct, type **y** to continue.



📌 **If there's an issue, type n to go back and reconfigure.**

Step 13: Finalizing Installation

1. First Boot Initialization

- The system will format **the second disk (IDE 1, 40GB)**.
- ClearPass will take a few minutes to initialize:

2. Login to ClearPass

- After booting, **log in using**:

Username: appadmin

Password: <your new password>

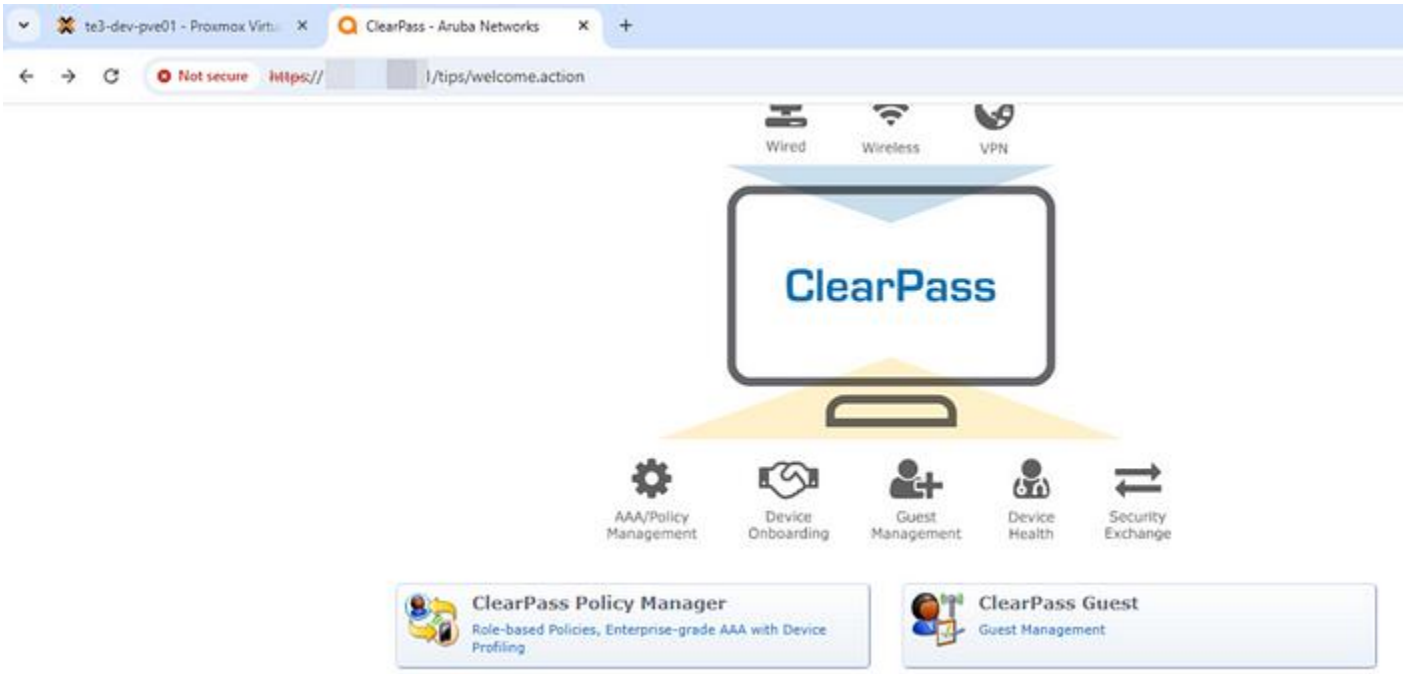
📌 **If you forgot the password, you'll need to reset it from the Proxmox console.**

✅ **Congratulations! Your ClearPass 6.12 VM is now fully installed and configured.** 🎉

Next Steps

🔹 **Access the ClearPass Web UI**: Open a browser and go to https://<management IP> , apply Aruba Clearpass platform license.

🔹 **Start configuring network policies**.



📌**If you cannot access the management IP, you may check the extra tipS below.**

💡 Extra Tips: Identifying the Correct Management Port in ClearPass

In some installations, the **management port may not always be assigned to net0** in Proxmox. Instead, it might be on **net1 or another interface**.

📌This may happen due to **interface detection order: Interfaces map based on the MAC address: *lower MAC address becomes the management port, and the higher MAC address becomes the data port.***

How to Identify the Management Port

1. **Log into ClearPass via CLI** and run "***show ip"***

- This will display the **IP addresses and MAC addresses** of all network interfaces. **Check the MAC address** of the interface labeled **Management Port** in ClearPass.

Example Screenshot (ClearPass CLI output):

```
[appadmin@CPPM-DEMO]# show ip

================================================
Device Type                    :   Management Port
------------------------------------------------
IPv4 Address                   :
Subnet Mask                    :
Gateway                        :

IPv6 Address/Prefix Length    :   <not configured>
Gateway                        :   <not configured>

Hardware Address               :   bc:24:11:8a:80:9c
MTU                            :   1500
================================================
Device Type                    :   Data Port
------------------------------------------------
IPv4 Address                   :   <not configured>
Subnet Mask                    :   <not configured>
Gateway                        :   <not configured>

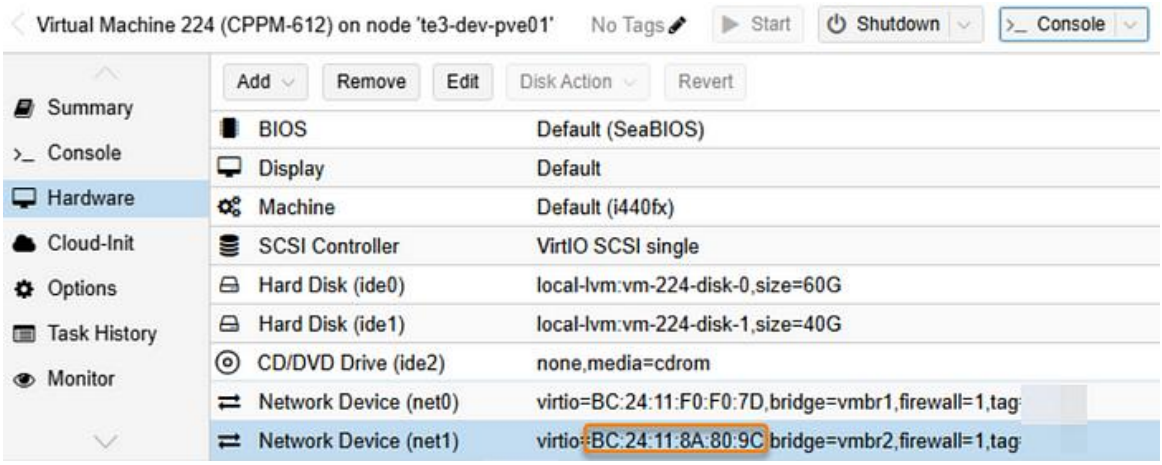
IPv6 Address/Prefix Length    :   <not configured>
Gateway                        :   <not configured>

Hardware Address               :   bc:24:11:f0:f0:7d
MTU                            :   1500
```

2. Verify the MAC address in Proxmox

- Go to **Proxmox Web UI**
- Open **Hardware → Network Devices**
- Compare the MAC addresses listed for **net0, net1, etc.**

Example Screenshot (Proxmox Network Configuration):



3. **Set up the correct management network**

- in this example, **MAC address of the management port** matches **net1(lower MAC address)** instead of **net0**, so you need to configure your Proxmox network settings accordingly. Ensure **net1** is connected to the correct bridge for management traffic.

Congrats, you have done the installation!

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